

codata

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NAME

codata – libray for fundamental physical constants

LIBRARY

Codata library (*libcodata*, *-lcodata*)

SYNOPSIS

Fortran **use codata**

C **#include "codata.h"**

Python **import pycodata**

DESCRIPTION

codata is a Fortran library providing the fundamental physical constants published by the CODATA. It includes the 2022, 2018, 2014 and 2010 recommended values, allowing applications to reproduce calculations based on specific CODATA releases.

The constants are implemented as a derived type that carries: *the name, the value, the uncertainty, and the unit*. All the constants are provided as **double precision reals**. Long constant names can be aliased with shorter names.

It provides:

- CODATA 2022 constants
- CODATA 2018 constants
- CODATA 2014 constants
- CODATA 2010 constants
- Native Fortran API
- C interoperability API
- Python bindings

Fortran API

exposes the constants as derived type **codata_constant_type** which carries *the name, the value, the uncertainty, and the unit*.

C API

exposes a structure **codata_constant_type** that defines the same members as in Fortran i.e. *the name, the value, the uncertainty, and the unit*.

Python

exposes a dictionary with the keys being *the name, the value, the uncertainty, and the unit*.

NOTES

The latest values (2022) do not have the year as a suffix in their name whereas older values (2010, 2014, 2018) feature the year as a suffix. All constants are declared with the same variable name in Fortran, C, and Python.

- [1] P. J. Mohr, B. N. Taylor, and D. B. Newell,
CODATA Recommended Values of the Fundamental Physical Constants: 2010,
Review of Modern Physics, vol. 84, 2012.
- [2] P. J. Mohr, B. N. Taylor, and D. B. Newell,
CODATA Recommended Values of the Fundamental Physical Constants: 2014,
Journal of Physical and Chemical Reference Data, vol. 45, 2016.
- [3] P. J. Mohr, B. N. Taylor, and D. B. Newell,
CODATA Recommended Values of the Fundamental Physical Constants: 2018,
Review of Modern Physics, vol. 93, 2021.
- [4] P. Mohr, D. Newell, B. Taylor, and E. Tiesinga,
CODATA Recommended Values of the Fundamental Physical Constants: 2022,
Accessed: May 05, 2025. [Online]. Available:
<https://arxiv.org/abs/2409.03787>
- [5] P. J. Mohr, D. B. Newell, B. N. Taylor, and E. Tiesinga,
CODATA Recommended Values of the Fundamental Physical Constants: 2022,
Reviews of Modern Physics, vol. 97, no. 2, p. 25002, Apr. 2025,
doi: 10.1103/RevModPhys.97.025002.

EXAMPLE

Example in Fortran

```

program example_in_f
use codata
implicit none
print '(A)', '##### EXAMPLE IN FORTRAN #####'
print '(A)', '# VERSION'
print *, "version = ", version()
print '(A)', '# CONSTANTS'
print *, "c = ", SPEED_OF_LIGHT_IN_VACUUM%value
print '(A)', '# UNCERTAINTY'
print *, "u(c) = ", SPEED_OF_LIGHT_IN_VACUUM%uncertainty
print '(A)', '# OLDER VALUES'
print '(A, F23.16)', "Mu_2022(latest) = ", MOLAR_MASS_CONSTANT%value
print '(A, F23.16)', "Mu_2018 = ", MOLAR_MASS_CONSTANT_2018%value
print '(A, F23.16)', "Mu_2014 = ", MOLAR_MASS_CONSTANT_2014%value
print '(A, F23.16)', "Mu_2010 = ", MOLAR_MASS_CONSTANT_2010%value
end program

```

Example in C

```

#include <stdio.h>
#include "codata.h"

int main(void){
printf("%f", c);
printf("##### EXAMPLE IN C #####\n");
printf("%s\n", "# VERSION");
printf("version = %s\n", codata_version());
printf("%s\n", "# CONSTANTS");
printf("c = %f\n", SPEED_OF_LIGHT_IN_VACUUM.value);
printf("%s\n", "# UNCERTAINTY");
printf("u(c) = %f\n", SPEED_OF_LIGHT_IN_VACUUM.uncertainty);
printf("%s\n", "# OLDER VALUES");
printf("Mu_2022(latest) = %23.16f\n", MOLAR_MASS_CONSTANT.value);
printf("Mu_2018 = %23.16f\n", MOLAR_MASS_CONSTANT_2018.value);
}

```

```
printf("Mu_2014 = %23.16f\n", MOLAR_MASS_CONSTANT_2014.value);
printf("Mu_2010 = %23.16f\n", MOLAR_MASS_CONSTANT_2010.value);
return 0;
}
```

Example in Python

```
import sys
sys.path.insert(0, "../py/src/")
import pycodata
print("##### EXAMPLE IN PYTHON #####")
print("# VERSION")
print(f"version = {pycodata.__version__}")
print("# Constants")
print("c =", pycodata.SPEED_OF_LIGHT_IN_VACUUM["value"])
print("# UNCERTAINTY")
print("u(c) = ", pycodata.SPEED_OF_LIGHT_IN_VACUUM["uncertainty"])
print("# OLDER VALUES")
print("Mu_2022 = ", pycodata.MOLAR_MASS_CONSTANT["value"])
print("Mu_2018 = ", pycodata.MOLAR_MASS_CONSTANT_2018["value"])
print("Mu_2014 = ", pycodata.MOLAR_MASS_CONSTANT_2014["value"])
print("Mu_2010 = ", pycodata.MOLAR_MASS_CONSTANT_2010["value"])
```

SEE ALSO

codata(3), codata_cli(1), codata_version(3), codata_constant_type(3)

CODATA 2022

List of available constants:

- ALPHA_PARTICLE_ELECTRON_MASS_RATIO
- ALPHA_PARTICLE_MASS
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_IN_MEV
- ALPHA_PARTICLE_MASS_IN_U
- ALPHA_PARTICLE_MOLAR_MASS
- ALPHA_PARTICLE_PROTON_MASS_RATIO
- ALPHA_PARTICLE_RELATIVE_ATOMIC_MASS
- ALPHA_PARTICLE_RMS_CHARGE_RADIUS
- ANGSTROM_STAR
- ATOMIC_MASS_CONSTANT
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_IN_MEV
- ATOMIC_MASS_UNIT_ELECTRON_VOLT_RELATIONSHIP
- ATOMIC_MASS_UNIT_HARTREE_RELATIONSHIP
- ATOMIC_MASS_UNIT_HERTZ_RELATIONSHIP

- ATOMIC_MASS_UNIT_INVERSE_METER_RELATIONSHIP
- ATOMIC_MASS_UNIT_JOULE_RELATIONSHIP
- ATOMIC_MASS_UNIT_KELVIN_RELATIONSHIP
- ATOMIC_MASS_UNIT_KILOGRAM_RELATIONSHIP
- ATOMIC_UNIT_OF_1ST_HYPERPOLARIZABILITY
- ATOMIC_UNIT_OF_2ND_HYPERPOLARIZABILITY
- ATOMIC_UNIT_OF_ACTION
- ATOMIC_UNIT_OF_CHARGE
- ATOMIC_UNIT_OF_CHARGE_DENSITY
- ATOMIC_UNIT_OF_CURRENT
- ATOMIC_UNIT_OF_ELECTRIC_DIPOLE_MOM
- ATOMIC_UNIT_OF_ELECTRIC_FIELD
- ATOMIC_UNIT_OF_ELECTRIC_FIELD_GRADIENT
- ATOMIC_UNIT_OF_ELECTRIC_POLARIZABILITY
- ATOMIC_UNIT_OF_ELECTRIC_POTENTIAL
- ATOMIC_UNIT_OF_ELECTRIC_QUADRUPOLE_MOM
- ATOMIC_UNIT_OF_ENERGY
- ATOMIC_UNIT_OF_FORCE
- ATOMIC_UNIT_OF_LENGTH
- ATOMIC_UNIT_OF_MAG_DIPOLE_MOM
- ATOMIC_UNIT_OF_MAG_FLUX_DENSITY
- ATOMIC_UNIT_OF_MAGNETIZABILITY
- ATOMIC_UNIT_OF_MASS
- ATOMIC_UNIT_OF_MOMENTUM
- ATOMIC_UNIT_OF_PERMITTIVITY
- ATOMIC_UNIT_OF_TIME
- ATOMIC_UNIT_OF_VELOCITY
- AVOGADRO_CONSTANT
- BOHR_MAGNETON
- BOHR_MAGNETON_IN_EV_T
- BOHR_MAGNETON_IN_HZ_T
- BOHR_MAGNETON_IN_INVERSE_METER_PER_TESLA
- BOHR_MAGNETON_IN_K_T
- BOHR_RADIUS
- BOLTZMANN_CONSTANT
- BOLTZMANN_CONSTANT_IN_EV_K
- BOLTZMANN_CONSTANT_IN_HZ_K
- BOLTZMANN_CONSTANT_IN_INVERSE_METER_PER_KELVIN

- CHARACTERISTIC_IMPEDANCE_OF_VACUUM
- CLASSICAL_ELECTRON_RADIUS
- COMPTON_WAVELENGTH
- CONDUCTANCE_QUANTUM
- CONVENTIONAL_VALUE_OF_AMPERE_90
- CONVENTIONAL_VALUE_OF_COULOMB_90
- CONVENTIONAL_VALUE_OF_FARAD_90
- CONVENTIONAL_VALUE_OF_HENRY_90
- CONVENTIONAL_VALUE_OF_JOSEPHSON_CONSTANT
- CONVENTIONAL_VALUE_OF_OHM_90
- CONVENTIONAL_VALUE_OF_VOLT_90
- CONVENTIONAL_VALUE_OF_VON_KLITZING_CONSTANT
- CONVENTIONAL_VALUE_OF_WATT_90
- COPPER_X_UNIT
- DEUTERON_ELECTRON_MAG_MOM_RATIO
- DEUTERON_ELECTRON_MASS_RATIO
- DEUTERON_G_FACTOR
- DEUTERON_MAG_MOM
- DEUTERON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- DEUTERON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- DEUTERON_MASS
- DEUTERON_MASS_ENERGY_EQUIVALENT
- DEUTERON_MASS_ENERGY_EQUIVALENT_IN_MEV
- DEUTERON_MASS_IN_U
- DEUTERON_MOLAR_MASS
- DEUTERON_NEUTRON_MAG_MOM_RATIO
- DEUTERON_PROTON_MAG_MOM_RATIO
- DEUTERON_PROTON_MASS_RATIO
- DEUTERON_RELATIVE_ATOMIC_MASS
- DEUTERON_RMS_CHARGE_RADIUS
- ELECTRON_CHARGE_TO_MASS_QUOTIENT
- ELECTRON_DEUTERON_MAG_MOM_RATIO
- ELECTRON_DEUTERON_MASS_RATIO
- ELECTRON_G_FACTOR
- ELECTRON_GYROMAG_RATIO
- ELECTRON_GYROMAG_RATIO_IN_MHZ_T
- ELECTRON_HELION_MASS_RATIO
- ELECTRON_MAG_MOM

- ELECTRON_MAG_MOM_ANOMALY
- ELECTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- ELECTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- ELECTRON_MASS
- ELECTRON_MASS_ENERGY_EQUIVALENT
- ELECTRON_MASS_ENERGY_EQUIVALENT_IN_MEV
- ELECTRON_MASS_IN_U
- ELECTRON_MOLAR_MASS
- ELECTRON_MUON_MAG_MOM_RATIO
- ELECTRON_MUON_MASS_RATIO
- ELECTRON_NEUTRON_MAG_MOM_RATIO
- ELECTRON_NEUTRON_MASS_RATIO
- ELECTRON_PROTON_MAG_MOM_RATIO
- ELECTRON_PROTON_MASS_RATIO
- ELECTRON_RELATIVE_ATOMIC_MASS
- ELECTRON_TAU_MASS_RATIO
- ELECTRON_TO_ALPHA_PARTICLE_MASS_RATIO
- ELECTRON_TO_SHIELDED_HELION_MAG_MOM_RATIO
- ELECTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO
- ELECTRON_TRITON_MASS_RATIO
- ELECTRON_VOLT
- ELECTRON_VOLT_ATOMIC_MASS_UNIT_RELATIONSHIP
- ELECTRON_VOLT_HARTREE_RELATIONSHIP
- ELECTRON_VOLT_HERTZ_RELATIONSHIP
- ELECTRON_VOLT_INVERSE_METER_RELATIONSHIP
- ELECTRON_VOLT_JOULE_RELATIONSHIP
- ELECTRON_VOLT_KELVIN_RELATIONSHIP
- ELECTRON_VOLT_KILOGRAM_RELATIONSHIP
- ELEMENTARY_CHARGE
- ELEMENTARY_CHARGE_OVER_H_BAR
- FARADAY_CONSTANT
- FERMI_COUPLING_CONSTANT
- FINE_STRUCTURE_CONSTANT
- FIRST_RADIATION_CONSTANT
- FIRST_RADIATION_CONSTANT_FOR_SPECTRAL_RADIANCE
- HARTREE_ATOMIC_MASS_UNIT_RELATIONSHIP
- HARTREE_ELECTRON_VOLT_RELATIONSHIP
- HARTREE_ENERGY

- HARTREE_ENERGY_IN_EV
- HARTREE_HERTZ_RELATIONSHIP
- HARTREE_INVERSE_METER_RELATIONSHIP
- HARTREE_JOULE_RELATIONSHIP
- HARTREE_KELVIN_RELATIONSHIP
- HARTREE_KILOGRAM_RELATIONSHIP
- HELION_ELECTRON_MASS_RATIO
- HELION_G_FACTOR
- HELION_MAG_MOM
- HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- HELION_MASS
- HELION_MASS_ENERGY_EQUIVALENT
- HELION_MASS_ENERGY_EQUIVALENT_IN_MEV
- HELION_MASS_IN_U
- HELION_MOLAR_MASS
- HELION_PROTON_MASS_RATIO
- HELION_RELATIVE_ATOMIC_MASS
- HELION_SHIELDING_SHIFT
- HERTZ_ATOMIC_MASS_UNIT_RELATIONSHIP
- HERTZ_ELECTRON_VOLT_RELATIONSHIP
- HERTZ_HARTREE_RELATIONSHIP
- HERTZ_INVERSE_METER_RELATIONSHIP
- HERTZ_JOULE_RELATIONSHIP
- HERTZ_KELVIN_RELATIONSHIP
- HERTZ_KILOGRAM_RELATIONSHIP
- HYPERFINE_TRANSITION_FREQUENCY_OF_CS_133
- INVERSE_FINE_STRUCTURE_CONSTANT
- INVERSE_METER_ATOMIC_MASS_UNIT_RELATIONSHIP
- INVERSE_METER_ELECTRON_VOLT_RELATIONSHIP
- INVERSE_METER_HARTREE_RELATIONSHIP
- INVERSE_METER_HERTZ_RELATIONSHIP
- INVERSE_METER_JOULE_RELATIONSHIP
- INVERSE_METER_KELVIN_RELATIONSHIP
- INVERSE_METER_KILOGRAM_RELATIONSHIP
- INVERSE_OF_CONDUCTANCE_QUANTUM
- JOSEPHSON_CONSTANT
- JOULE_ATOMIC_MASS_UNIT_RELATIONSHIP

- JOULE_ELECTRON_VOLT_RELATIONSHIP
- JOULE_HARTREE_RELATIONSHIP
- JOULE_HERTZ_RELATIONSHIP
- JOULE_INVERSE_METER_RELATIONSHIP
- JOULE_KELVIN_RELATIONSHIP
- JOULE_KILOGRAM_RELATIONSHIP
- KELVIN_ATOMIC_MASS_UNIT_RELATIONSHIP
- KELVIN_ELECTRON_VOLT_RELATIONSHIP
- KELVIN_HARTREE_RELATIONSHIP
- KELVIN_HERTZ_RELATIONSHIP
- KELVIN_INVERSE_METER_RELATIONSHIP
- KELVIN_JOULE_RELATIONSHIP
- KELVIN_KILOGRAM_RELATIONSHIP
- KILOGRAM_ATOMIC_MASS_UNIT_RELATIONSHIP
- KILOGRAM_ELECTRON_VOLT_RELATIONSHIP
- KILOGRAM_HARTREE_RELATIONSHIP
- KILOGRAM_HERTZ_RELATIONSHIP
- KILOGRAM_INVERSE_METER_RELATIONSHIP
- KILOGRAM_JOULE_RELATIONSHIP
- KILOGRAM_KELVIN_RELATIONSHIP
- LATTICE_PARAMETER_OF_SILICON
- LATTICE_SPACING_OF_IDEAL_SI_220
- LOSCHMIDT_CONSTANT_273_15_K_100_KPA
- LOSCHMIDT_CONSTANT_273_15_K_101_325_KPA
- LUMINOUS EFFICACY
- MAG_FLUX_QUANTUM
- MOLAR_GAS_CONSTANT
- MOLAR_MASS_CONSTANT
- MOLAR_MASS_OF_CARBON_12
- MOLAR_PLANCK_CONSTANT
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_100_KPA
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_101_325_KPA
- MOLAR_VOLUME_OF_SILICON
- MOLYBDENUM_X_UNIT
- MUON_COMPTON_WAVELENGTH
- MUON_ELECTRON_MASS_RATIO
- MUON_G_FACTOR
- MUON_MAG_MOM

- MUON_MAG_MOM_ANOMALY
- MUON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- MUON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- MUON_MASS
- MUON_MASS_ENERGY_EQUIVALENT
- MUON_MASS_ENERGY_EQUIVALENT_IN_MEV
- MUON_MASS_IN_U
- MUON_MOLAR_MASS
- MUON_NEUTRON_MASS_RATIO
- MUON_PROTON_MAG_MOM_RATIO
- MUON_PROTON_MASS_RATIO
- MUON_TAU_MASS_RATIO
- NATURAL_UNIT_OF_ACTION
- NATURAL_UNIT_OF_ACTION_IN_EV_S
- NATURAL_UNIT_OF_ENERGY
- NATURAL_UNIT_OF_ENERGY_IN_MEV
- NATURAL_UNIT_OF_LENGTH
- NATURAL_UNIT_OF_MASS
- NATURAL_UNIT_OF_MOMENTUM
- NATURAL_UNIT_OF_MOMENTUM_IN_MEV_C
- NATURAL_UNIT_OF_TIME
- NATURAL_UNIT_OF_VELOCITY
- NEUTRON_COMPTON_WAVELENGTH
- NEUTRON_ELECTRON_MAG_MOM_RATIO
- NEUTRON_ELECTRON_MASS_RATIO
- NEUTRON_G_FACTOR
- NEUTRON_GYROMAG_RATIO
- NEUTRON_GYROMAG_RATIO_IN_MHZ_T
- NEUTRON_MAG_MOM
- NEUTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- NEUTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- NEUTRON_MASS
- NEUTRON_MASS_ENERGY_EQUIVALENT
- NEUTRON_MASS_ENERGY_EQUIVALENT_IN_MEV
- NEUTRON_MASS_IN_U
- NEUTRON_MOLAR_MASS
- NEUTRON_MUON_MASS_RATIO
- NEUTRON_PROTON_MAG_MOM_RATIO

- NEUTRON_PROTON_MASS_DIFFERENCE
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_IN_MEV
- NEUTRON_PROTON_MASS_DIFFERENCE_IN_U
- NEUTRON_PROTON_MASS_RATIO
- NEUTRON_RELATIVE_ATOMIC_MASS
- NEUTRON_TAU_MASS_RATIO
- NEUTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO
- NEWTONIAN_CONSTANT_OF_GRAVITATION
- NEWTONIAN_CONSTANT_OF_GRAVITATION_OVER_H_BAR_C
- NUCLEAR_MAGNETON
- NUCLEAR_MAGNETON_IN_EV_T
- NUCLEAR_MAGNETON_IN_INVERSE_METER_PER_TESLA
- NUCLEAR_MAGNETON_IN_K_T
- NUCLEAR_MAGNETON_IN_MHZ_T
- PLANCK_CONSTANT
- PLANCK_CONSTANT_IN_EV_HZ
- PLANCK_LENGTH
- PLANCK_MASS
- PLANCK_MASS_ENERGY_EQUIVALENT_IN_GEV
- PLANCK_TEMPERATURE
- PLANCK_TIME
- PROTON_CHARGE_TO_MASS_QUOTIENT
- PROTON_COMPTON_WAVELENGTH
- PROTON_ELECTRON_MASS_RATIO
- PROTON_G_FACTOR
- PROTON_GYROMAG_RATIO
- PROTON_GYROMAG_RATIO_IN_MHZ_T
- PROTON_MAG_MOM
- PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- PROTON_MAG_SHIELDING_CORRECTION
- PROTON_MASS
- PROTON_MASS_ENERGY_EQUIVALENT
- PROTON_MASS_ENERGY_EQUIVALENT_IN_MEV
- PROTON_MASS_IN_U
- PROTON_MOLAR_MASS
- PROTON_MUON_MASS_RATIO

- PROTON_NEUTRON_MAG_MOM_RATIO
- PROTON_NEUTRON_MASS_RATIO
- PROTON_RELATIVE_ATOMIC_MASS
- PROTON_RMS_CHARGE_RADIUS
- PROTON_TAU_MASS_RATIO
- QUANTUM_OF_CIRCULATION
- QUANTUM_OF_CIRCULATION_TIMES_2
- REDUCED_COMPTON_WAVELENGTH
- REDUCED_MUON_COMPTON_WAVELENGTH
- REDUCED_NEUTRON_COMPTON_WAVELENGTH
- REDUCED_PLANCK_CONSTANT
- REDUCED_PLANCK_CONSTANT_IN_EV_S
- REDUCED_PLANCK_CONSTANT_TIMES_C_IN_MEV_FM
- REDUCED_PROTON_COMPTON_WAVELENGTH
- REDUCED_TAU_COMPTON_WAVELENGTH
- RYDBERG_CONSTANT
- RYDBERG_CONSTANT_TIMES_C_IN_HZ
- RYDBERG_CONSTANT_TIMES_HC_IN_EV
- RYDBERG_CONSTANT_TIMES_HC_IN_J
- SACKUR_TETRODE_CONSTANT_1_K_100_KPA
- SACKUR_TETRODE_CONSTANT_1_K_101_325_KPA
- SECOND_RADIATION_CONSTANT
- SHIELDED_HELION_GYROMAG_RATIO
- SHIELDED_HELION_GYROMAG_RATIO_IN_MHZ_T
- SHIELDED_HELION_MAG_MOM
- SHIELDED_HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- SHIELDED_HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- SHIELDED_HELION_TO_PROTON_MAG_MOM_RATIO
- SHIELDED_HELION_TO_SHIELDED_PROTON_MAG_MOM_RATIO
- SHIELDED_PROTON_GYROMAG_RATIO
- SHIELDED_PROTON_GYROMAG_RATIO_IN_MHZ_T
- SHIELDED_PROTON_MAG_MOM
- SHIELDED_PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- SHIELDED_PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- SHIELDING_DIFFERENCE_OF_D_AND_P_IN_HD
- SHIELDING_DIFFERENCE_OF_T_AND_P_IN_HT
- SPEED_OF_LIGHT_IN_VACUUM
- STANDARD_ACCELERATION_OF_GRAVITY

- STANDARD_ATMOSPHERE
- STANDARD_STATE_PRESSURE
- STEFAN_BOLTZMANN_CONSTANT
- TAU_COMPTON_WAVELENGTH
- TAU_ELECTRON_MASS_RATIO
- TAU_ENERGY_EQUIVALENT
- TAU_MASS
- TAU_MASS_ENERGY_EQUIVALENT
- TAU_MASS_IN_U
- TAU_MOLAR_MASS
- TAU_MUON_MASS_RATIO
- TAU_NEUTRON_MASS_RATIO
- TAU_PROTON_MASS_RATIO
- THOMSON_CROSS_SECTION
- TRITON_ELECTRON_MASS_RATIO
- TRITON_G_FACTOR
- TRITON_MAG_MOM
- TRITON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- TRITON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- TRITON_MASS
- TRITON_MASS_ENERGY_EQUIVALENT
- TRITON_MASS_ENERGY_EQUIVALENT_IN_MEV
- TRITON_MASS_IN_U
- TRITON_MOLAR_MASS
- TRITON_PROTON_MASS_RATIO
- TRITON_RELATIVE_ATOMIC_MASS
- TRITON_TO_PROTON_MAG_MOM_RATIO
- UNIFIED_ATOMIC_MASS_UNIT
- VACUUM_ELECTRIC_PERMITTIVITY
- VACUUM_MAG_PERMEABILITY
- VON_KLITZING_CONSTANT
- WEAK_MIXING_ANGLE
- WIEN_FREQUENCY_DISPLACEMENT_LAW_CONSTANT
- WIEN_WAVELENGTH_DISPLACEMENT_LAW_CONSTANT
- W_TO_Z_MASS_RATIO

CODATA 2018

List of available constants:

- ALPHA_PARTICLE_ELECTRON_MASS_RATIO _2018

- ALPHA_PARTICLE_MASS_2018
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_2018
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- ALPHA_PARTICLE_MASS_IN_U_2018
- ALPHA_PARTICLE_MOLAR_MASS_2018
- ALPHA_PARTICLE_PROTON_MASS_RATIO_2018
- ALPHA_PARTICLE_RELATIVE_ATOMIC_MASS_2018
- ANGSTROM_STAR_2018
- ATOMIC_MASS_CONSTANT_2018
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_2018
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_IN_MEV_2018
- ATOMIC_MASS_UNIT_ELECTRON_VOLT_RELATIONSHIP_2018
- ATOMIC_MASS_UNIT_HARTREE_RELATIONSHIP_2018
- ATOMIC_MASS_UNIT_HERTZ_RELATIONSHIP_2018
- ATOMIC_MASS_UNIT_INVERSE_METER_RELATIONSHIP_2018
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- LOSCHMIDT_CONSTANT_273_15_K_100_KPA_2010
- LOSCHMIDT_CONSTANT_273_15_K_101_325_KPA_2010
- MAG_CONSTANT_2010

- MAG_FLUX_QUANTUM _2010
- MOLAR_GAS_CONSTANT _2010
- MOLAR_MASS_CONSTANT _2010
- MOLAR_MASS_OF_CARBON_12 _2010
- MOLAR_PLANCK_CONSTANT _2010
- MOLAR_PLANCK_CONSTANT_TIMES_C _2010
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_100_KPA _2010
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_101_325_KPA _2010
- MOLAR_VOLUME_OF_SILICON _2010
- MO_X_UNIT _2010
- MUON_COMPTON_WAVELENGTH _2010
- MUON_COMPTON_WAVELENGTH_OVER_2_PI _2010
- MUON_ELECTRON_MASS_RATIO _2010
- MUON_G_FACTOR _2010
- MUON_MAG_MOM _2010
- MUON_MAG_MOM_ANOMALY _2010
- MUON_MAG_MOM_TO_BOHR_MAGNETON_RATIO _2010
- MUON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO _2010
- MUON_MASS _2010
- MUON_MASS_ENERGY_EQUIVALENT _2010
- MUON_MASS_ENERGY_EQUIVALENT_IN_MEV _2010
- MUON_MASS_IN_U _2010
- MUON_MOLAR_MASS _2010
- MUON_NEUTRON_MASS_RATIO _2010
- MUON_PROTON_MAG_MOM_RATIO _2010
- MUON_PROTON_MASS_RATIO _2010
- MUON_TAU_MASS_RATIO _2010
- NATURAL_UNIT_OF_ACTION _2010
- NATURAL_UNIT_OF_ACTION_IN_EV_S _2010
- NATURAL_UNIT_OF_ENERGY _2010
- NATURAL_UNIT_OF_ENERGY_IN_MEV _2010
- NATURAL_UNIT_OF_LENGTH _2010
- NATURAL_UNIT_OF_MASS _2010
- NATURAL_UNIT_OF_MOMUM _2010
- NATURAL_UNIT_OF_MOMUM_IN_MEV_C _2010
- NATURAL_UNIT_OF_TIME _2010
- NATURAL_UNIT_OF_VELOCITY _2010
- NEUTRON_COMPTON_WAVELENGTH _2010

- NEUTRON_COMPTON_WAVELENGTH_OVER_2_PI_2010
- NEUTRON_ELECTRON_MAG_MOM_RATIO_2010
- NEUTRON_ELECTRON_MASS_RATIO_2010
- NEUTRON_G_FACTOR_2010
- NEUTRON_GYROMAG_RATIO_2010
- NEUTRON_GYROMAG_RATIO_OVER_2_PI_2010
- NEUTRON_MAG_MOM_2010
- NEUTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- NEUTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- NEUTRON_MASS_2010
- NEUTRON_MASS_ENERGY_EQUIVALENT_2010
- NEUTRON_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- NEUTRON_MASS_IN_U_2010
- NEUTRON_MOLAR_MASS_2010
- NEUTRON_MUON_MASS_RATIO_2010
- NEUTRON_PROTON_MAG_MOM_RATIO_2010
- NEUTRON_PROTON_MASS_DIFFERENCE_2010
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_2010
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_IN_MEV_2010
- NEUTRON_PROTON_MASS_DIFFERENCE_IN_U_2010
- NEUTRON_PROTON_MASS_RATIO_2010
- NEUTRON_TAU_MASS_RATIO_2010
- NEUTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2010
- NEWTONIAN_CONSTANT_OF_GRAVITATION_2010
- NEWTONIAN_CONSTANT_OF_GRAVITATION_OVER_H_BAR_C_2010
- NUCLEAR_MAGNETON_2010
- NUCLEAR_MAGNETON_IN_EV_T_2010
- NUCLEAR_MAGNETON_IN_INVERSE_METERS_PER_TESLA_2010
- NUCLEAR_MAGNETON_IN_K_T_2010
- NUCLEAR_MAGNETON_IN_MHZ_T_2010
- PLANCK_CONSTANT_2010
- PLANCK_CONSTANT_IN_EV_S_2010
- PLANCK_CONSTANT_OVER_2_PI_2010
- PLANCK_CONSTANT_OVER_2_PI_IN_EV_S_2010
- PLANCK_CONSTANT_OVER_2_PI_TIMES_C_IN_MEV_FM_2010
- PLANCK_LENGTH_2010
- PLANCK_MASS_2010
- PLANCK_MASS_ENERGY_EQUIVALENT_IN_GEV_2010

- PLANCK_TEMPERATURE_2010
- PLANCK_TIME_2010
- PROTON_CHARGE_TO_MASS_QUOTIENT_2010
- PROTON_COMPTON_WAVELENGTH_2010
- PROTON_COMPTON_WAVELENGTH_OVER_2_PI_2010
- PROTON_ELECTRON_MASS_RATIO_2010
- PROTON_G_FACTOR_2010
- PROTON_GYROMAG_RATIO_2010
- PROTON_GYROMAG_RATIO_OVER_2_PI_2010
- PROTON_MAG_MOM_2010
- PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- PROTON_MAG_SHIELDING_CORRECTION_2010
- PROTON_MASS_2010
- PROTON_MASS_ENERGY_EQUIVALENT_2010
- PROTON_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- PROTON_MASS_IN_U_2010
- PROTON_MOLAR_MASS_2010
- PROTON_MUON_MASS_RATIO_2010
- PROTON_NEUTRON_MAG_MOM_RATIO_2010
- PROTON_NEUTRON_MASS_RATIO_2010
- PROTON_RMS_CHARGE_RADIUS_2010
- PROTON_TAU_MASS_RATIO_2010
- QUANTUM_OF_CIRCULATION_2010
- QUANTUM_OF_CIRCULATION_TIMES_2_2010
- RYDBERG_CONSTANT_2010
- RYDBERG_CONSTANT_TIMES_C_IN_HZ_2010
- RYDBERG_CONSTANT_TIMES_HC_IN_EV_2010
- RYDBERG_CONSTANT_TIMES_HC_IN_J_2010
- SACKUR_TETRODE_CONSTANT_1_K_100_KPA_2010
- SACKUR_TETRODE_CONSTANT_1_K_101_325_KPA_2010
- SECOND_RADIATION_CONSTANT_2010
- SHIELDED_HELION_GYROMAG_RATIO_2010
- SHIELDED_HELION_GYROMAG_RATIO_OVER_2_PI_2010
- SHIELDED_HELION_MAG_MOM_2010
- SHIELDED_HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- SHIELDED_HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- SHIELDED_HELION_TO_PROTON_MAG_MOM_RATIO_2010

- SHIELDED_HELION_TO_SHIELDED_PROTON_MAG_MOM_RATIO _2010
- SHIELDED_PROTON_GYROMAG_RATIO _2010
- SHIELDED_PROTON_GYROMAG_RATIO_OVER_2_PI _2010
- SHIELDED_PROTON_MAG_MOM _2010
- SHIELDED_PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO _2010
- SHIELDED_PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO _2010
- SPEED_OF_LIGHT_IN_VACUUM _2010
- STANDARD_ACCELERATION_OF_GRAVITY _2010
- STANDARD_ATMOSPHERE _2010
- STANDARD_STATE_PRESSURE _2010
- STEFAN_BOLTZMANN_CONSTANT _2010
- TAU_COMPTON_WAVELENGTH _2010
- TAU_COMPTON_WAVELENGTH_OVER_2_PI _2010
- TAU_ELECTRON_MASS_RATIO _2010
- TAU_MASS _2010
- TAU_MASS_ENERGY_EQUIVALENT _2010
- TAU_MASS_ENERGY_EQUIVALENT_IN_MEV _2010
- TAU_MASS_IN_U _2010
- TAU_MOLAR_MASS _2010
- TAU_MUON_MASS_RATIO _2010
- TAU_NEUTRON_MASS_RATIO _2010
- TAU_PROTON_MASS_RATIO _2010
- THOMSON_CROSS_SECTION _2010
- TRITON_ELECTRON_MASS_RATIO _2010
- TRITON_G_FACTOR _2010
- TRITON_MAG_MOM _2010
- TRITON_MAG_MOM_TO_BOHR_MAGNETON_RATIO _2010
- TRITON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO _2010
- TRITON_MASS _2010
- TRITON_MASS_ENERGY_EQUIVALENT _2010
- TRITON_MASS_ENERGY_EQUIVALENT_IN_MEV _2010
- TRITON_MASS_IN_U _2010
- TRITON_MOLAR_MASS _2010
- TRITON_PROTON_MASS_RATIO _2010
- UNIFIED_ATOMIC_MASS_UNIT _2010
- VON_KLITZING_CONSTANT _2010
- WEAK_MIXING_ANGLE _2010
- WIEN_FREQUENCY_DISPLACEMENT_LAW_CONSTANT _2010

- WIEN_WAVELENGTH_DISPLACEMENT_LAW_CONSTANT_2010

NAME

codata – libray for fundamental physical constants

SYNOPSIS

codata [*OPTION*]...

DESCRIPTION

codata is a command line interface which prints all the CODATA constants.

The current values are from 2022. Older values can be retrieved if needed and the output can be filtered with *REGEX PATTERNS*.

OPTIONS

- y, --year *YEAR***
CODATA constants: 2022, 2018, 2014, 2010
- p, --pattern *PATTERN***
Regex pattern for filtering the constants.
- a, --value**
Show only the value.
- e, --error**
Show only the uncertainty.
- u, --usage**
Show usage text and exit.
- v, --version**
Show version information and exit.
- h, --help**
Show help text and exit.

EXAMPLE

```
Minimal example
codata
codata -p Faraday
codata -y 2014 -p molar.*gas,electron.*eV
codata [B,b]oltzmann.*eV
```

SEE ALSO

codata(3), codata_cli(1), codata_version(3), codata_constant_type(3)

NAME

version – function

LIBRARYCodata library (*libcodata*, *-lcodata*)**SYNOPSIS**

Fortran **function version(result(*fptr*)**
 character(len=:), pointer :: *fptr*

C **char* ciaaw_version()**

Python **str:pyciaaw.__version__**

DESCRIPTION

Get the library version.

RETURN VALUE

pointer Pointer to the version string.

EXAMPLE

Fortran:

```
use codata
print *, "version = ", version()
```

C:

```
include <stdio.h>
include "codata.h"
printf("version = %s\n", codata_version());
```

Python:

```
import pycodata
print(f"version = {pycodata.__version__}")
```

SEE ALSO**codata(3), codata_cli(1), codata_version(3), codata_constant_type(3)**

NAME

codata_constant_type – derived type

LIBRARY

Codata library (*libcodata*, *-lcodata*)

SYNOPSIS

Fortran **type(codata_constant_type) :: c**

C **struct codata_constant_type c**

DESCRIPTION

Defines a constant with *the name, the value, the uncertainty and the unit*.

```

type :: codata_constant_type
    character(len=64) :: name
    real(dp) :: value
    real(dp) :: uncertainty
    character(len=32) :: unit
contains
    procedure :: print
    procedure :: to_real_sp
    procedure :: to_real_dp
    generic :: to_real => to_real_sp, to_real_dp
end type codata_constant_type

struct codata_constant_type{
    char name[65];
    double value;
    double uncertainty;
    char unit[33];
};

```

EXAMPLE

Fortran:

```

use codata
type(codata_constant_type), pointer :: c
c = SPEED_OF_LIGHT_IN_VACUUM

```

C:

```

include "codata.h"
struct codata_constant_type c;
c = SPEED_OF_LIGHT_IN_VACUUM;

```

SEE ALSO**codata(3), codata_cli(1), codata_version(3), codata_constant_type(3)**